Enhancing Cognition and Academic Performance in Children Through Exergaming and Dual-Task Training

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Introduction

Cognitive development and academic success in children are heavily influenced by executive functions (EFs), which encompass cognitive flexibility, inhibitory control, and attention. Recent research suggests that incorporating physical activity into cognitive training, particularly through exergaming and dual-task training (DTT), offers substantial benefits for cognitive performance and academic achievement. Exergaming, a form of physical activity that integrates interactive video gaming, and DTT, which requires individuals to engage in simultaneous cognitive and motor tasks, have both been associated with improvements in executive function, postural control, and learning outcomes (Kou et al., 2024; Wollesen et al., 2022).

The Impact of Exergaming on Cognition

Exergaming has been shown to enhance various executive functions, including cognitive flexibility, inhibitory control, and global cognition. A systematic review and meta-analysis conducted by Kou et al. (2024) found that exergaming significantly improved cognitive flexibility (SMD = 0.34), inhibitory control (SMD = 0.51), and global cognition (SMD = 0.87) in children. These cognitive enhancements translate into better academic performance, as executive functions play a critical role in attentional control, working memory, and problem-solving abilities. Additionally, studies have highlighted the positive effects of cognitively engaging movement games on academic achievement. Chou et al. (2023) demonstrated that movement-based cognitive tasks significantly improved interference control, a crucial EF for learning, in overweight children. Similarly, Kolovelonis et al. (2023) found that acute exergaming sessions in elementary physical education classes enhanced students' executive functions and situational interest in learning activities.

Dual-Task Training and Cognitive Development

Dual-task training (DTT) involves performing cognitive and motor tasks simultaneously, which has been shown to improve both cognitive and motor performance. A scoping review by Wollesen et al. (2022) highlighted that DTT interventions enhance executive functions and motor coordination in children and adolescents. These benefits are particularly notable in populations with cognitive and motor deficits, such as adolescents with Down syndrome. Triki et al. (2024) found that an eight-week DTT program significantly improved both postural control and cognitive performance in adolescents with Down syndrome, demonstrating the intervention's potential for enhancing functional abilities. Furthermore, Bustillo-Casero et al. (2020) reported that DTT led to greater improvements in dual-task conditions compared to single-task

training (STT), reinforcing the importance of integrated cognitive-motor training.

Age-Related Differences in Dual-Task Training Benefits

Research suggests that older children benefit more from dual-task training due to their more developed executive functions. Möhring et al. (2021) found that children aged 8-13 demonstrated better dual-task performance than younger children, particularly in updating and switching tasks. Similarly, Cowan et al. (2021) observed that children aged 10-14 showed greater attention allocation and cognitive control when balancing two simultaneous tasks compared to children aged 6-8. Yang et al. (2017) documented a continuous linear improvement in multitasking abilities from ages 7 to 12, indicating that as children age, their cognitive and motor skills develop further, making them more adept at dual-tasking.

Training Duration and Frequency

The duration and frequency of dual-task training significantly impact cognitive outcomes in children, particularly in enhancing executive functions and academic performance. Longer interventions tend to yield greater cognitive benefits, with studies showing that a nine-week coordination training program incorporating mental tasks resulted in significant improvements in both motor and cognitive functions in children aged 6-10 (Santner et al., 2018). Additionally, an eight-week DTT intervention improved postural and cognitive performance in adolescents with Down syndrome (Triki et al., 2024). Chen et al. (2021) conducted a meta-analysis on cognitive interventions in children with ADHD and found that higher training frequency was associated with greater executive function improvements. Similarly, Johann and Karbach (2020) demonstrated that 21 sessions of executive control training led to significant gains in academic performance, particularly in reading and mathematical skills.

Comparing Dual-Task Training to Traditional Cognitive Training

While traditional cognitive training methods focus primarily on cognitive enhancements, dual-task training offers a more holistic approach by integrating cognitive and motor skills. DTT has been found to be more effective in improving executive functions and academic performance than STT (Wollesen et al., 2022). Triki et al. (2024) demonstrated that dual-task interventions produced greater improvements in postural control and cognitive performance than traditional cognitive exercises. Bustillo-Casero et al. (2020) reported that adolescents participating in DTT exhibited superior motor-cognitive integration, which is critical for real-world problem-solving. On the other hand, traditional cognitive training methods, such as game-based executive control training, have also shown benefits. Johann and Karbach (2020) found that game-based cognitive training increased motivation and domain-specific cognitive improvements, particularly in reading performance. However, dual-task training provides a broader range of benefits by simultaneously enhancing motor coordination and cognitive functions,

making it a more effective strategy for holistic cognitive development.

Conclusion

Exergaming and dual-task training offer promising interventions for enhancing cognition and academic performance in children. By integrating physical activity with cognitive engagement, these approaches improve executive functions, including cognitive flexibility, inhibitory control, and attention. Older children and adolescents benefit the most from DTT, as their executive functions are more developed. Additionally, longer and more frequent training sessions yield greater cognitive improvements. While traditional cognitive training methods remain beneficial, dual-task training provides more comprehensive benefits by enhancing both cognitive and motor skills, making it a superior approach for academic success. Future educational programs should incorporate these strategies to optimize cognitive development in children.

References

Bustillo-Casero, P., Cebrian-Bou, S., Cruz-Montecinos, C., Pardo, A., & García-Massó, X. (2020). Effects of a dual-task intervention in postural control and cognitive performance in adolescents. *Journal of Motor Behavior, 52*(2), 187-195. https://doi.org/10.1080/00222895.2019.1600467

Chen, S., Yu, J., Zhang, Q., et al. (2021). Which factor is more relevant to the effectiveness of the cognitive intervention? A meta-analysis of randomized controlled trials of cognitive training on symptoms and executive function behaviors of children with attention deficit hyperactivity disorder. *Frontiers in Psychology, 12*, 810298. https://doi.org/10.3389/fpsyg.2021.810298

Chou, C. C., Kao, S. C., Pan, C. C., et al. (2023). Cognitively engaging movement games improve interference control and academic performance in overweight children: A randomized control trial. *Scandinavian Journal of Medicine & Science in Sports, 33*(4), 521-534. https://doi.org/10.1111/sms.14264

Johann, V. E., & Karbach, J. (2020). Effects of game-based and standard executive control training on cognitive and academic abilities in elementary school children. *Developmental Science, 23*(4), e12866. https://doi.org/10.1111/desc.12866

Kolovelonis, A., Papastergiou, M., Samara, E., & Goudas, M. (2023). Acute effects of exergaming on students' executive functions and situational interest in elementary physical education. *International Journal of Environmental Research and Public Health, 20*(3), 1902. https://doi.org/10.3390/ijerph20031902

Kou, R., Zhang, Z., Zhu, F., Tang, Y., & Li, Z. (2024). Effects of exergaming on executive function and motor ability in children: A systematic review and meta-analysis. *PloS One, 19*(9),

e0309462. https://doi.org/10.1371/journal.pone.0309462

Triki, A., Borji, R., Laatar, R., Sahli, S., & Rebai, H. (2024). The effect of dual-task training on postural and cognitive performances in adolescents with Down syndrome. *Research in Developmental Disabilities, 153*, 104827. https://doi.org/10.1016/j.ridd.2024.104827

Wollesen, B., Janssen, T. I., Müller, H., & Voelcker-Rehage, C. (2022). Effects of cognitive-motor dual task training on cognitive and physical performance in healthy children and adolescents: A scoping review. *Acta Psychologica, 224*, 103498. https://doi.org/10.1016/j.actpsy.2022.103498